Teacher sheet lesson seven

Introduction

In the previous lesson you learned about formation of new crust at diverging plate boundaries. In this lesson you will examine destruction of old crust at converging plates.

Task 1 — Convergent plate boundaries

Student information:

Go to the website *Subduction zones* (http://www.platetectonics.com/book/page_12.asp) and read about convergent plate boundaries. Open a new Word document, create a labelled diagram of a subduction zone and include events and landforms that occur in this zone. Save this in the folder for this unit.

a) Why do volcanoes form at convergent plate boundaries?

b) Why do some places seem to have lots of earthquakes?

Task 2 — Subducting plate

Student information:

How do we know that one plate is sliding under another? Construct a model using the information provided in ‘Model instructions’. The model is based on data from earthquakes. Record your group’s activity and submit to your teacher. View another group’s recording.

Teacher information:

For more information about the subducting plate model, follow this link.

http://www.geosociety.org/educate/LessonPlans/RealEvidence-SubductingPlate.pdf

Reflection questions and instructions for this activity are included below.

Model instructions

The model you will be making today draws on actual earthquake data to provide evidence for the existence of a subducting plate. Earthquakes occur deep inside the Earth and the depth and size of all earthquakes is continuously being recorded.

Research the answers to the following questions to review your understanding of earthquakes.

1. What is an earthquake?
2. Where do earthquakes usually occur?
3. What is the difference between the epicentre and the focus of an earthquake?

Form into groups of four. Follow the instructions below to construct a model that shows the shape of the Nazca plate subducting under the South American plate. Remember you will need to make a recording of the activity and save it in the folder for this unit.

First you need to collect the following materials from the front of the room.

- Box
- 22 balls (or beads) of three colours (whichever your teacher has provided)
- Glue
- Masking tape
- 2 rulers
- Scissors
- Copy of the map
- Copy of the earthquake data table
- 22 paper clips
- String

1. Use the information in the table to plot the earthquakes on the map provided.
2. Lay the box on the side and glue the map onto the top-facing side of the box. The bottom of the map should face the open end of the box.
3. At each epicentre on the map punch a small hole. It only needs to be big enough for the string to fit through.
4. Each of the balls (beads) will represent the focus of the earthquake. Each focus was recorded at its particular depth and these depths are contained within the table. For each epicentre of the map, find its focus. You will then need to attach a ball to a length of string that matches the depth of the focus. Make sure that you use an appropriate scale or you will not see the pattern in your finished model. Start with the deepest earthquake so you know they will all fit.
5. Work through all 22 epicentres on the map and hang each ball at the depth of each focus.
6. When you are finished with your model, answer the following questions either in a Word document or verbally as part of your recorded presentation.

Questions

1. What information does an epicentre provide about an earthquake?
2. What is the difference between the focus and the epicentre of an earthquake?
3. What is happening to the rocks at the focus of an earthquake, and what do we call this?
4. What do you notice about the depth of the focus of the earthquakes as you go further inland from the coast of South America?
5. According to your model, what appears to be happening to the two plates that meet along the west coast of South America?
6. Draw and label a diagram of the model showing what is happening to the plates along the west coast of South America as described in question 5.
7. Describe the type of plate boundary that is present along the west coast of South America.
8. Explain how your data supports your conclusions in question 5.
9. How can your model explain the deep trench that lies off the coast of South America?
10. Use a map of the world that shows the plate boundaries to find some other areas where this same type of plate boundary is occurring.
11. How is volcanism related to this type of boundary?